



Indian Institute of Information Technology Vadodara

Mid-Semester-Examination-Autumn-2025-26

Program: MCA

CS445: Digital Design and Computer Architecture

MM: 30

Time: 2.0 Hours

Read the instructions carefully.

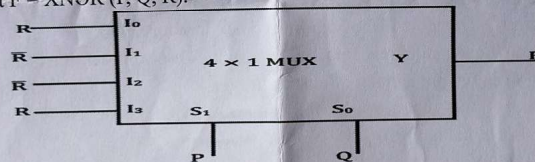
- Attempt ALL the questions.
- Make point wise answers for the required part.
- The calculator is not permitted in the examination hall.

Question 1: (1+1+1+1+1=05 Marks)

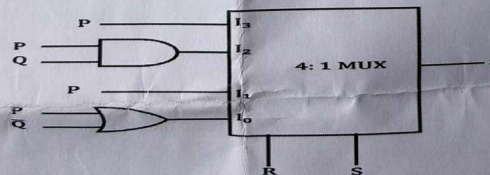
- (a) A function $F(A, B, C)$ defined by three Boolean variables A, B and C when expressed as sum of products is given by $F = A'B'C' + A'B'C + A \cdot B' \cdot C'$ where, A' , B' , and C' are the complements of the respective variables. Calculate the product of sums (POS) form of the function.
- (b) Minimize the expression $F(A, B, C) = AB + A'C + BC$ using Boolean theorems. *BC*
- (c) Perform a binary addition $A+B$ where, $A = 10011$ and $B = 11101$. *10000*
- (d) Minimize the following expression using K-map $F = A'C + A'B + AB'C + BC$. *C + A'B*
- (e) Derive the Boolean expression for the half subtractor (FA) circuit and implement HS circuit using 2x4 decoder circuit.

Question 2: (2.5+2.5 = 05 Marks)

- (a) The output F of the multiplexer circuit shown below expressed in terms of the inputs P, Q and R . Prove that $F = \text{XNOR}(P, Q, R)$.



- (b) For the circuit shown in the following figure, $I_0 - I_3$ are inputs to the 4:1 multiplexer. R (MSB) and S are control bits. Prove that $Z = PQ + PQ'S + Q'R'S'$.



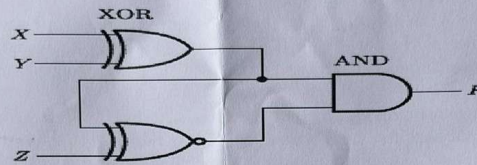
Question 3: (2.5+2.5 = 05 Marks)

- I. Derive the carry-lookahead expressions for C_1, C_2, C_3 , and C_4 in terms of G_i, P_i , and C_0 . Show algebraic steps.

- II. Implement the following POS function using $F = (X+Z)(Y'+Z)(X'+Y+Z)$. Take your phone number digits and add them if the sum is an odd number then realize this circuit using the two input NOR gates only else using the basic gates.

Question 4: (2.5+2.5 = 05 Marks)

- I. As we are aware that NAND and NOR gates are universal. Is a 4×1 MUX also universal, and can it be used to realize any logic function by proper selection and input assignment? If yes, show the implementation of NAND, NOR, XOR, and XNOR functions using 4×1 MUX.
- II. Determine the output expression F for the digital logic circuit shown in the figure below.



Question 5: (2.5+2.5 = 05 Marks)

- ✓1. Minimize the function using K-map, $F(x,y,z,w) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$. *$\bar{x}w + zw$*
- 0.5 II. Minimize the Boolean function $F(x,y,z,w) = \sum m(0,1,4,5,8,9,13,15)$ using a four-variable K-map, derive the final minimized SOP form with precise mapping of prime implicants to minterms. Transform the minimized expression into its canonical POS form, demonstrating each algebraic step and employing Boolean identities such as De Morgan's theorem where appropriate. Subsequently, design and implement two gate-level realizations of the function: (a) the minimized SOP using NAND logic, and (b) the minimized POS using OR-AND-NOT logic.

Question 6: (05 Marks)

A full adder is a fundamental combinational circuit that adds three binary inputs: two significant bits (A and B) and a carry input (Cin), producing a sum (S) and a carry output (Cout). The truth table of a full adder demonstrates that the outputs can be expressed in canonical sum-of-minterms form. A 3-to-8 line decoder is a versatile digital component that takes three binary inputs and generates eight mutually exclusive outputs, each corresponding to one of the minterms of the input variables. By properly combining the decoder outputs with external logic gates, any arbitrary Boolean function of three inputs can be implemented. Design a full adder using a single 3-to-8 decoder and the minimum number of additional gates required. Clearly explain how the decoder outputs correspond to the minterms of the full adder's truth table, and show how the sum (S) and carry (Cout) expressions can be realized through OR'ing the required decoder lines. Illustrate your solution with neat diagrams. Finally, discuss the importance of such decoders.

-----End of the Question Paper-----